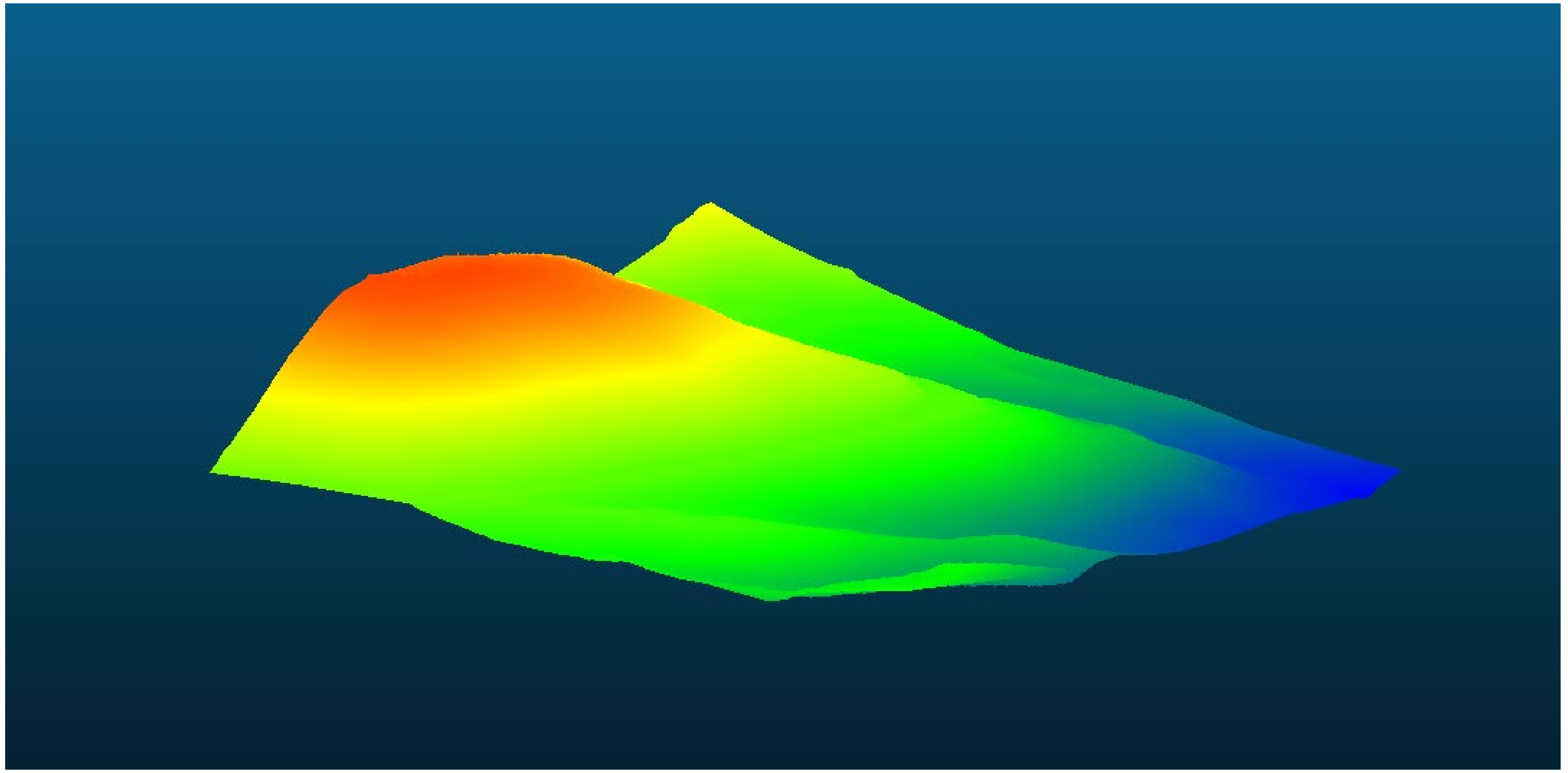
Max Larimer

Assignment 1

Cloud Compare

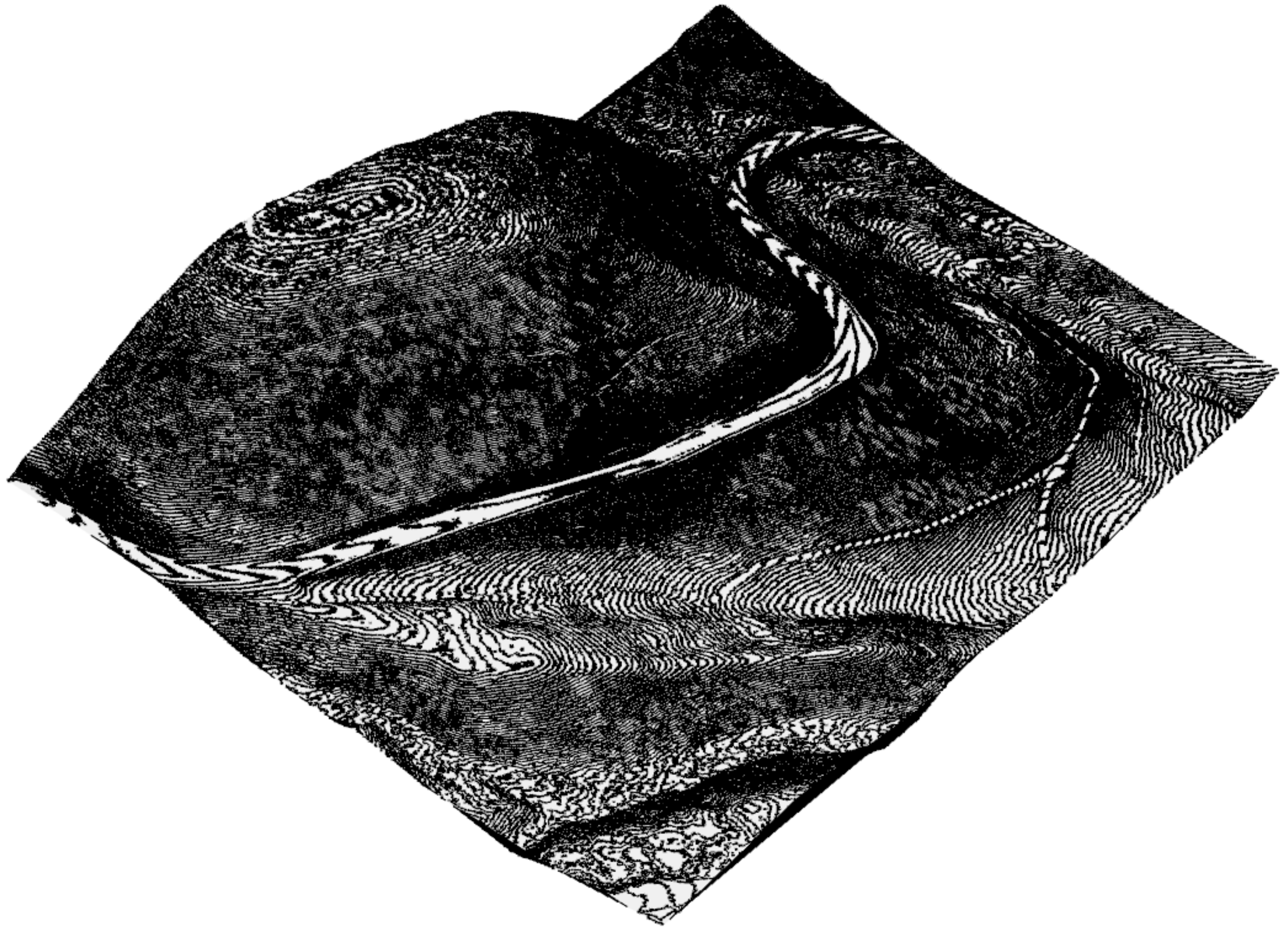
In the first stages of the project, we learned how to import and clean LiDAR data of topographic locations. Using the tool CloudCompare, we were able to separate off-ground points from ground points and create a working mesh that could be used as a project site. The LiDAR data was sourced from OpenTopography, and the project site is located in Big Sky, Montana along the road to the ski resort.

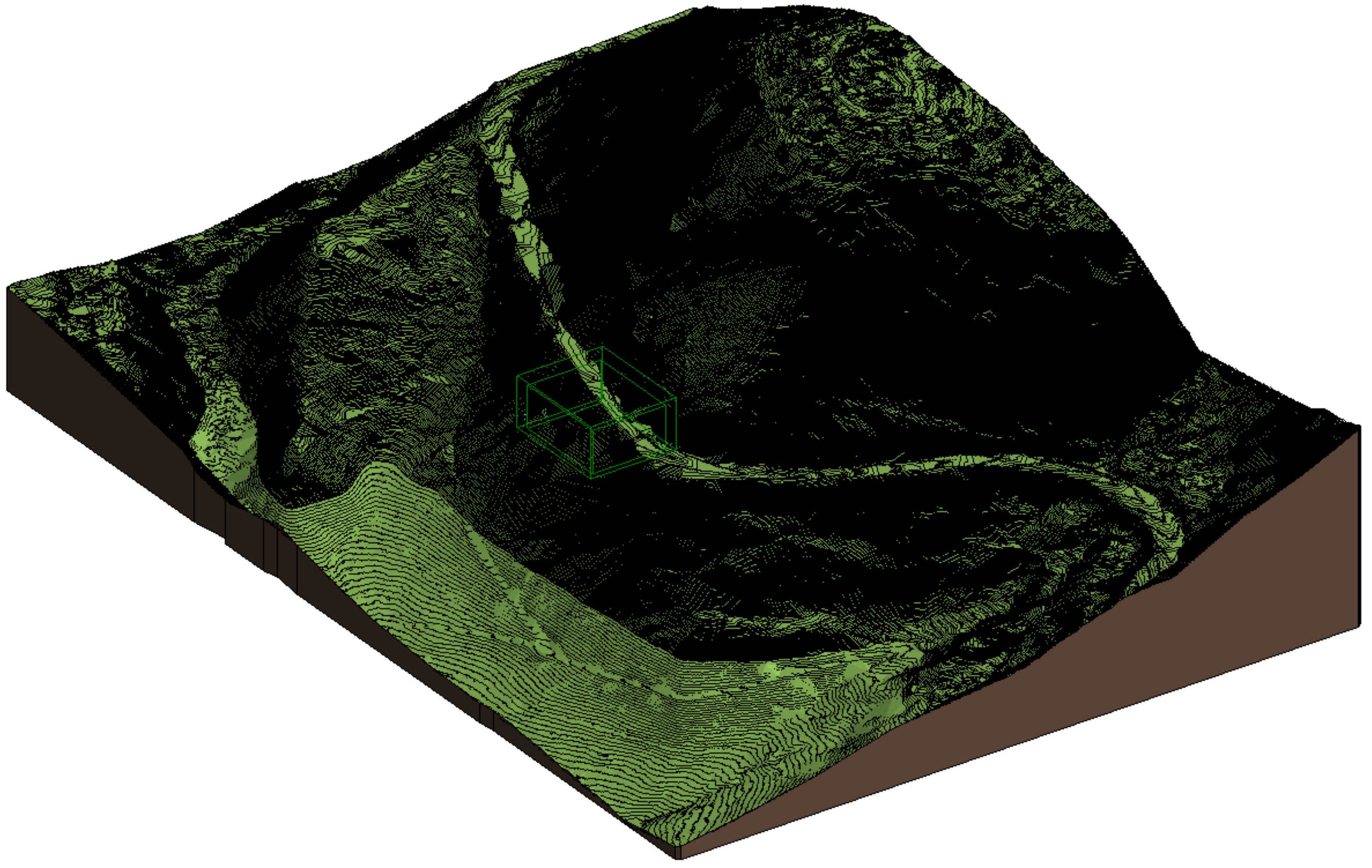


RHINO & REVIT

Contours - Topography

After exporting the mesh from CloudCompare, we learned how to create contoured meshes in Rhino and adjust the contour intervals. This then transferred into a Revit workflow, as we were able to export the contours as a DWG file and import them into Revit. Using the "Site from DWG" option, we were able to create a toposolid in Revit.

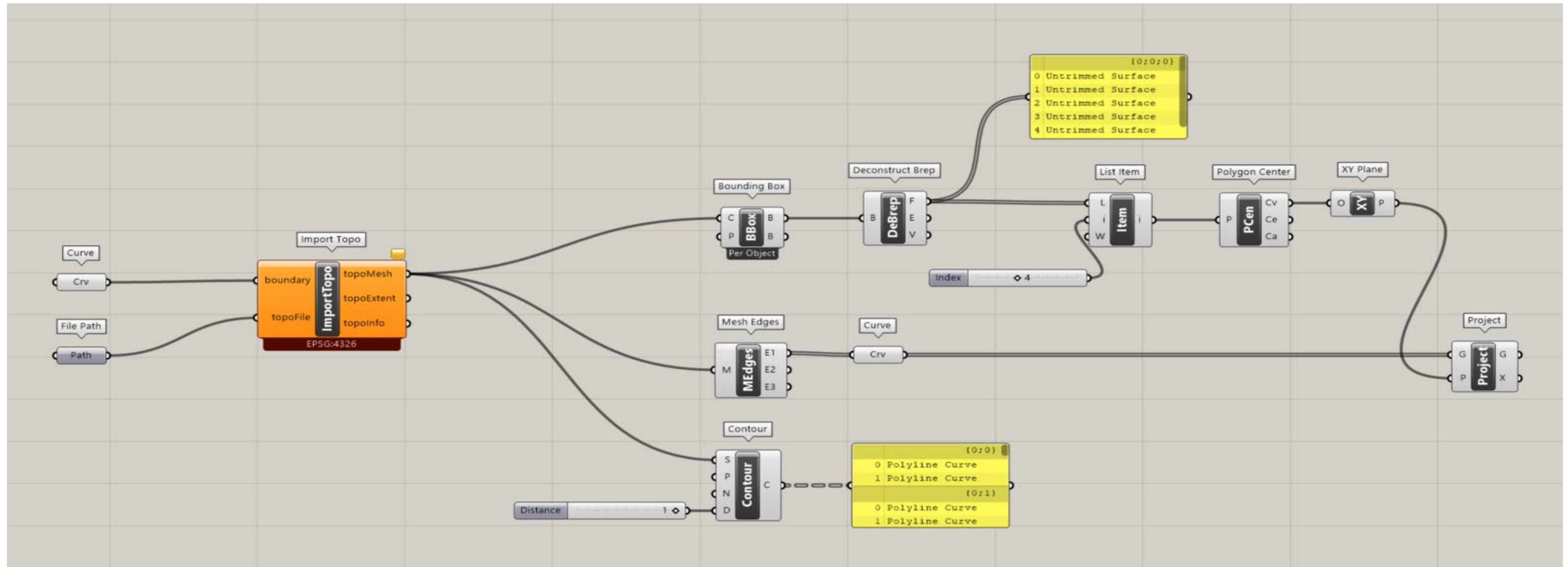


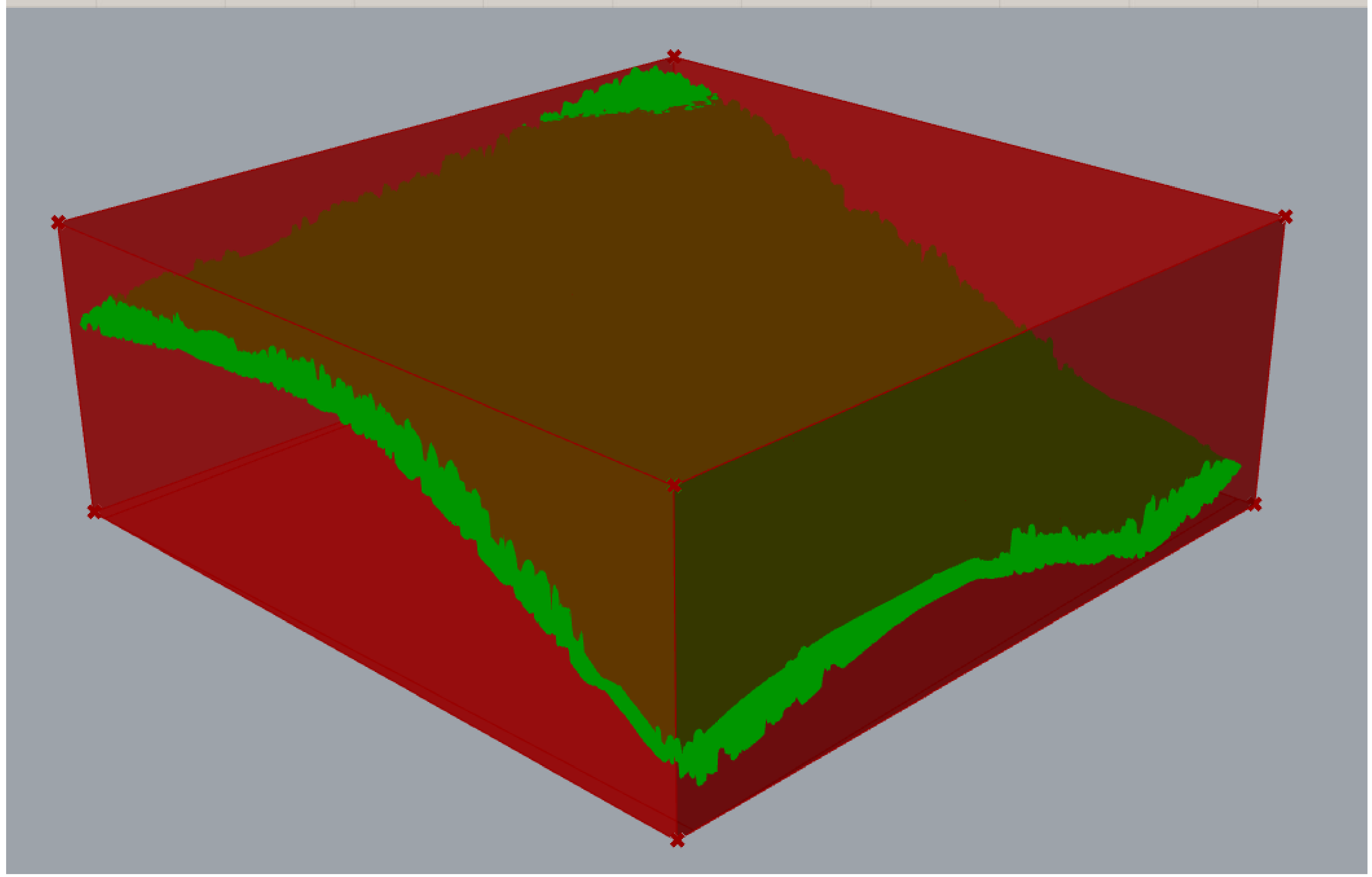


OSM

Contours & Starting Fabrication

In class, we learned another workflow for pulling topography data into Rhino. This process involved locating our site through OpenTopography and pulling data from OSM. When beginning the process of creating a laser cut file, we created bounding boxes, identified mesh edges, and sorted through list items to select the correct faces.

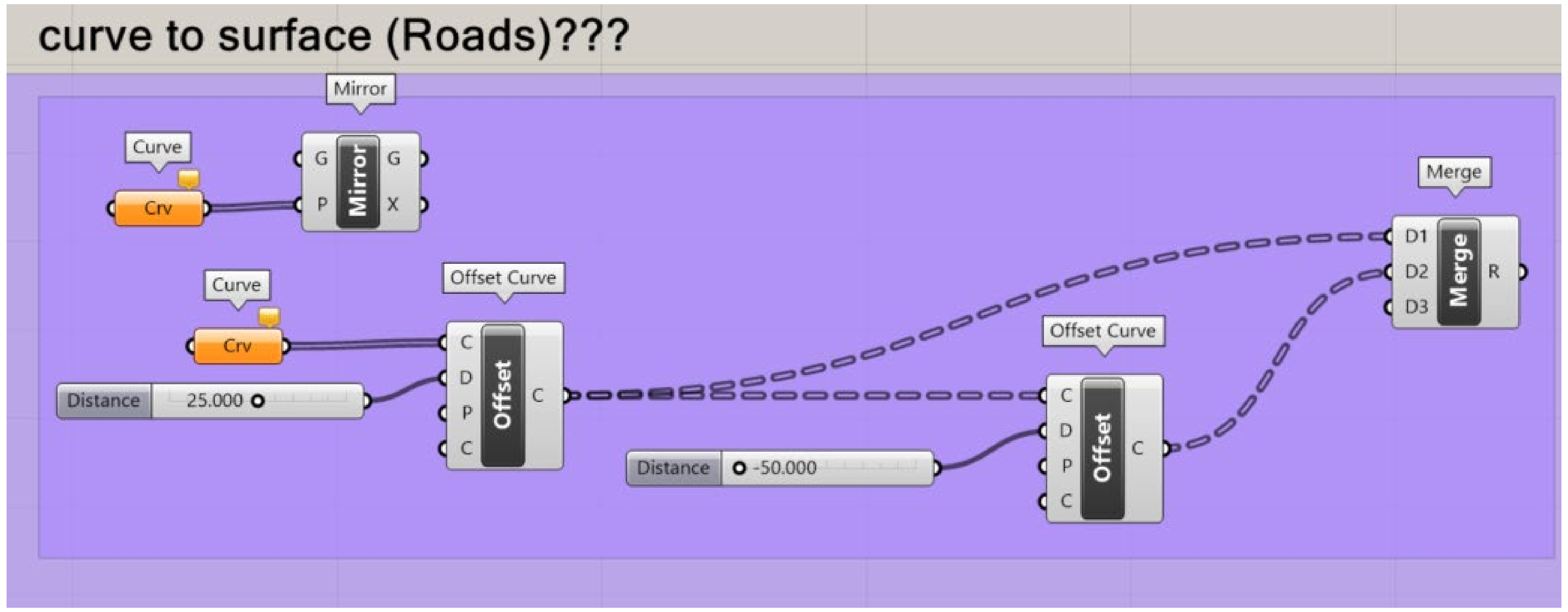


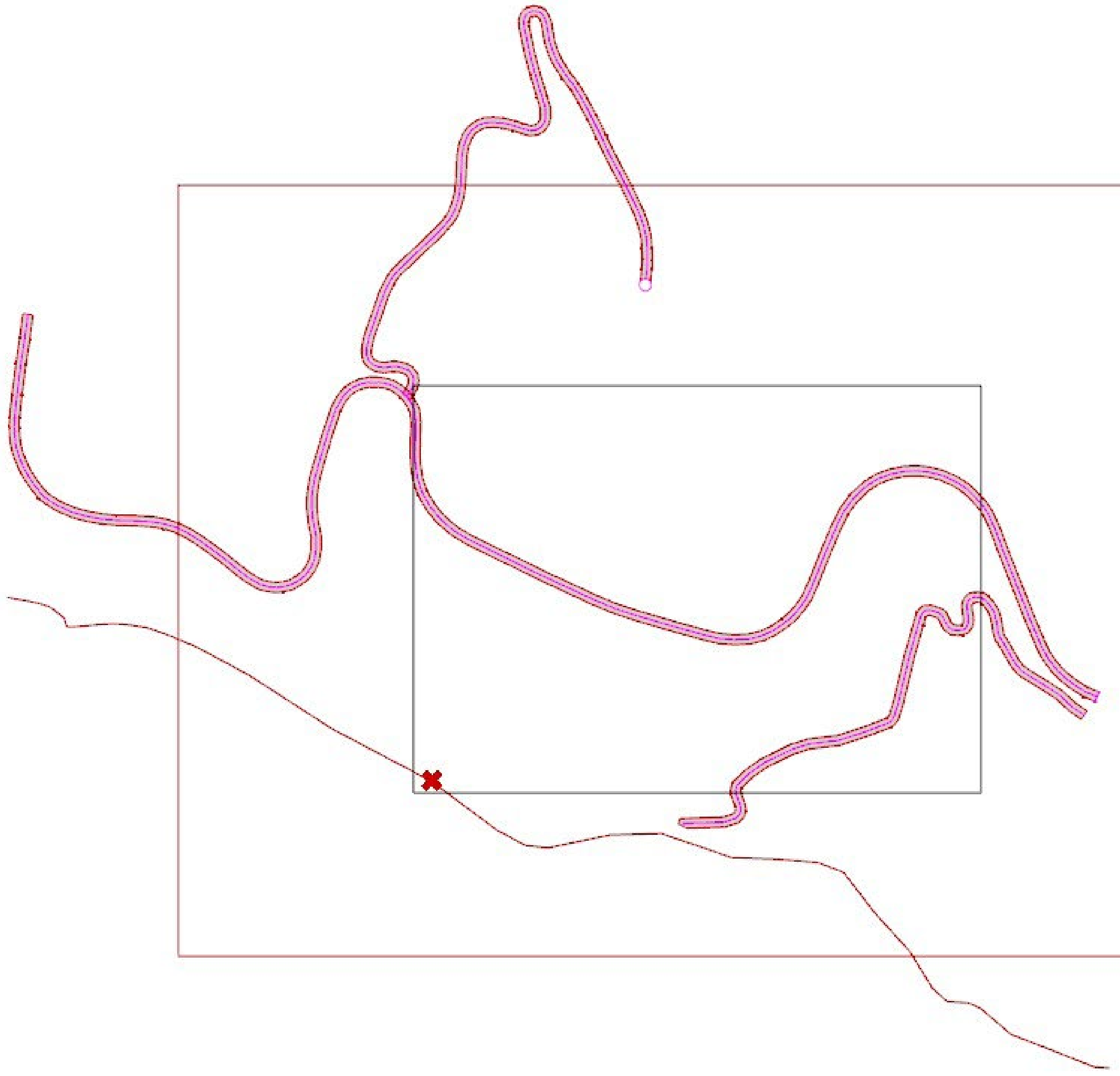


Roads

OSM DATA

Another step taken during the OSM data portion of the project involved adding more site context to our model. Logan and I pulled road data from OSM and created a curve. Using this curve, we attempted to assign a width to the road that could then be projected onto the mesh/topography.





Fabrication

Mesh - Laser cut template

In the final mesh to laser cut file workflow, we were provided with a full script that could take a mesh and convert it into a set of laser cut paths, allowing us to automate the file creation process. Logan and I ran into issues with our meshes producing open curves, and we attempted to create a fix using a combination of List Items, Extend Curves, and Rejoin. Through this process, we learned how to sort through data to identify problems within the script.